

Submersible Level Transmitter

Installation, Operation and Maintenance Manual

Document	Technical Manual	Edition	English
Revision	1.0	Product type	Gauge-pressure, vented

Specifications vary by ordered configuration. The product label, order specification and approved drawing take precedence over the typical values in this manual.

1. Product Overview

The submersible level transmitter measures liquid level from hydrostatic pressure. A silicon piezoresistive or ceramic sensing element converts the pressure at the submerged probe into an electrical signal. The signal is temperature-compensated, linearized and converted into a standard **2-wire 4–20 mA current output**.

The probe is installed directly in the liquid. The cable connects the probe to a controller, data logger, display or other receiving instrument. Typical applications include open tanks, wells, reservoirs, groundwater monitoring, water treatment systems and compatible industrial liquids.

Key Features

- Solid-state construction with no moving parts
- Direct submersible installation and continuous level measurement
- Temperature compensation and linearity correction
- 316 stainless-steel probe for compatible media
- Gauge-pressure measurement with atmospheric-pressure compensation through a vented cable
- Reverse-polarity and current-limiting protection, depending on configuration

IMPORTANT

This manual describes a gauge-pressure, vented transmitter. The cable vent provides the atmospheric reference required for compensation. Keep the vent path open, clean and dry. Blocking, sealing or flooding the vent can cause zero drift and weather-related measurement error.

2. Operating Principle

For a probe submerged to a vertical depth H below the liquid surface, the absolute pressure acting on the sensing diaphragm is the sum of hydrostatic pressure and atmospheric pressure:

$$P_{abs} = \rho \times g \times H + P_{atm}$$

Pabs	Absolute pressure at the sensing diaphragm
ρ	Density of the measured liquid
g	Local gravitational acceleration
H	Vertical liquid depth above the sensing diaphragm
Patm	Atmospheric pressure at the liquid surface

The reference side of the sensing element is connected to the surrounding atmosphere through the vent tube inside the cable. Atmospheric pressure therefore acts on both sides of the sensing element and is cancelled:

$$P_{gauge} = P_{abs} - P_{atm} = \rho \times g \times H$$

The liquid level is then calculated from the measured gauge pressure:

$$H = P_{\text{gauge}} / (\rho \times g)$$

PRINCIPLE	Atmospheric-pressure compensation is built into a correctly installed vented gauge-pressure transmitter. A separate barometric-pressure sensor is normally not required for an open tank, well or reservoir. This statement does not apply to sealed or pressurized vessels.
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Signal Conversion

The transmitter maps the calibrated pressure range to its electrical output. For a 4–20 mA version with a zero-to-full-scale level range, 4 mA represents zero level and 20 mA represents full scale:

$$I_{\text{out}} = 4 \text{ mA} + 16 \text{ mA} \times (H / \text{HFS})$$

For liquids other than water, use the actual operating density when converting pressure to level. A density change caused by temperature, concentration or composition produces a proportional level error unless the system compensates for it.

Application Limits

- Open or atmospherically vented vessels: use the gauge-pressure principle described above.
- Sealed or pressurized vessels: the gas-space pressure must be measured or referenced separately; a standard vented gauge transmitter alone cannot distinguish gas pressure from liquid head.
- Foam normally has little effect on hydrostatic pressure, but turbulence, direct jet impact, sediment and viscous deposits can create unstable or biased readings.

3. Typical Technical Specifications

Parameter	Typical value
Measurement range	1 kPa to 2 MPa, approximately 0.1 to 200 mH ₂ O; configurable
Pressure reference	Gauge pressure, vented
Accuracy class	0.1 / 0.25 / 0.5, depending on version
Long-term stability	≤ 0.1% FS per year
Supply voltage	12–36 VDC
Output options	4–20 mA, 2-wire
Probe material	316 stainless steel
Process-medium temperature	–30 to +70 °C
Ambient temperature	–40 to +85 °C
Ingress protection	IP68 for the submersible probe
Typical protection functions	Reverse-polarity and overcurrent/overvoltage protection, depending on version
NOTE	Do not infer certification, hazardous-area approval, chemical compatibility or load capacity from this general manual. Use only the ratings printed on the product label or stated in the approved product specification.

4. Installation

4.1 Before Installation

- Verify the ordered measurement range, supply voltage, process temperature, material compatibility and receiving-instrument scaling.
- Inspect the probe, cable and vent path for damage or blockage. Keep the transmitter in its original packaging in a cool, dry and ventilated location until installation.

4.2 Mechanical Installation

1. Install the probe vertically with the sensing diaphragm fully submerged. The effective level reference is the elevation of the sensing diaphragm, not the cable entry or enclosure.
2. Position the probe away from inlets, outlets, pumps, aerators and agitators. Direct flow impact creates dynamic pressure that is not liquid level.
3. Keep the probe above sediment and prevent it from contacting the tank or well bottom.
4. Support the transmitter with an approved cable clamp or separate suspension member where required. Do not lift, pull or repeatedly flex the cable at the probe entry.
5. Route the cable without sharp bends, crushing or abrasion. Keep the cable junction and vent termination above the maximum liquid level in a dry enclosure that still allows atmospheric breathing.

Stilling Well for Turbulent Liquids

For wells, tanks with strong circulation, or locations exposed to inflow, install the probe inside a stilling tube. A tube with an internal diameter of approximately 45 mm or larger is commonly suitable when it provides adequate clearance. Add openings at several heights so the liquid level inside the tube can equalize with the vessel. Place the openings away from the main flow direction and ensure they cannot become blocked.

DO NOT

Do not seal the cable vent, immerse the vent termination, fill the vent with adhesive, or place it in a permanently airtight enclosure. The vent is part of the measurement system, not an unused cable cavity.

4.3 Electrical Connection

Always follow the wiring diagram supplied with the ordered unit. Wire colors and pin assignments are configuration-specific.

- Disconnect power before wiring or inspection.
- For a 2-wire 4–20 mA loop, connect the transmitter in series with the DC supply and the receiving analog input.
- Observe polarity and confirm that the total loop resistance is within the permitted load for the selected supply voltage.
- Route signal cables separately from motors, contactors, variable-frequency drives and mains wiring.
- Where shielding is required, ground the shield at one end only unless the site grounding design specifies otherwise.

5. Commissioning and Scaling

1. Confirm the probe is installed at the intended zero reference and that the vent path is open and dry.
2. Apply the specified supply voltage and allow the signal to stabilize.
3. Check the zero output at zero applied gauge pressure. For a standard 4–20 mA range, the expected value is approximately 4 mA, subject to the ordered calibration and installation offset.
4. Immerse the probe to a known vertical depth and compare the measured value with the expected hydrostatic pressure or level.
5. Configure the controller or platform with the correct lower-range value, upper-range value, output span and engineering unit.
6. Enter the actual liquid density when the medium is not water or when the process requires high accuracy.

Example: 4–20 mA Scaling

For a transmitter calibrated for 0–10 m of liquid level, an output of 12 mA corresponds to 50% of span and therefore 5 m, provided that the liquid density matches the calibration basis:

$$\text{Level} = (12 - 4) / 16 \times 10 \text{ m} = 5 \text{ m}$$

6. Improving Measurement Stability

Unstable readings are usually caused by process dynamics or installation conditions rather than the sensing element. Correct the mechanical cause before increasing software filtering.

- Shield the probe from falling liquid and high-velocity jets.
- Use a stilling well or baffle to reduce wave action and dynamic pressure.
- Move the probe away from pump suction, discharge piping and agitators.
- Secure the cable so the probe does not swing or strike nearby structures.
- Use moderate averaging or a low-pass filter only after the installation is mechanically stable. Heavy filtering can hide real level changes and increase response delay.
- Check grounding, shielding and cable routing if the signal contains electrical noise.

7. Calibration

The transmitter is factory-calibrated to the ordered range and normally requires no field adjustment. Do not open a sealed IP68 probe body. Perform the following procedure only on a version with an approved, accessible zero/span adjustment module and only by qualified personnel.

1. Connect a regulated DC supply and a reference meter with suitable accuracy; 0.2% or better is recommended for checking common industrial accuracy classes.
2. Apply zero gauge pressure and adjust the zero output to 4 mA for a standard 4–20 mA configuration.
3. Apply the full-scale hydrostatic pressure and adjust the span output to 20 mA.
4. Repeat the zero and full-scale checks two or three times because the adjustments may interact slightly.
5. Verify the output at 25%, 50% and 75% of span.
6. When using a water column to calibrate a transmitter intended for another liquid, convert the target level by density. For example, a liquid with relative density 1.3 produces the same pressure at 1.0 m as approximately 1.3 m of water.
7. Restore all covers and seals after calibration and verify that the vent remains unobstructed.

A yearly calibration check is a practical default for many applications. Critical processes may require a shorter interval; stable, non-critical installations may use an interval established from historical calibration data and the site quality system.

8. Maintenance and Troubleshooting

Routine Maintenance

- Inspect the probe, cable, support point, junction and vent termination at planned intervals.
- Remove loose deposits with clean water and a soft brush. Do not scrape the diaphragm with metal tools or abrasive materials.
- Keep the vent path and any desiccant element dry and unobstructed.
- Check zero output after major maintenance, lightning events, cable damage or abnormal process conditions.

Troubleshooting Guide

Symptom	Check / corrective action
No output	Check supply voltage, polarity, loop continuity, fuse, input configuration and cable damage.

Symptom	Check / corrective action
Output fixed near minimum	Confirm that the probe is submerged, the measurement range is correct, and the receiving device is configured for a 4-20 mA current input .
Reading changes with weather	Inspect the atmospheric vent. A blocked, sealed or wet vent prevents correct barometric compensation.
Reading is unstable	Check for direct flow impact, wave action, pump turbulence, probe movement, electrical interference or poor grounding.
Constant offset	Verify the zero reference, probe elevation, vent condition and controller scaling.
Incorrect level for non-water liquid	Correct the density used in the pressure-to-level conversion.
Slow response	Inspect the stilling well and pressure port for blockage and review excessive digital filtering.

9. Safety and Handling

- Use the transmitter only within its specified voltage, pressure, temperature and material-compatibility limits.
- Do not use the product in a hazardous atmosphere unless the exact ordered version carries the required approval and is installed according to that approval.
- De-energize the circuit before installation, removal or wiring work.
- Stop using the transmitter if the cable, probe, sealing system or enclosure is damaged.
- Follow local electrical, environmental and process-safety requirements.

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